

A HYBRID DEEP LEARNING-BASED MULTIMODAL EXPLAINABLE MOVIE RECOMMENDATION SYSTEM

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of the requirements for the degree

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**INTEGRATED MASTER OF SCIENCE
IN
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by

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This is to certify that the dissertation entitled “**A HYBRID DEEP LEARNING-BASED MULTIMODAL EXPLAINABLE MOVIE RECOMMENDATION SYSTEM**” submitted by **Ms. MADHURA BASHINI M S** (Reg. No.: **CB.PS.I5DAS22032**), for the award of the **Degree of Integrated Master of Science in Data Science** is a bonafide record of the work carried out by her at the Department of Mathematics, Amrita School of Physical Sciences, Amrita Vishwa Vidyapeetham, Coimbatore.

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DECLARATION

I, **Ms. Madhura Bashini M S** (Register Number : CB.PS.I5DAS22032), hereby declare that this dissertation entitled “**A Hybrid Deep Learning-Based Multimodal Explainable Movie Recommendation System**” is the record of the original work done by me at the Department of Mathematics, Amrita School of Physical Sciences, Amrita Vishwa Vidyapeetham, Coimbatore. To the best of my knowledge this work has not formed the basis for the award of any degree/diploma/associateship/fellowship/or a similar award to any candidate in any University.

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(Madhura Bashini M S)

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List of Symbols

NCF	-	Neural Collaborative Filtering
SBERT	-	Sentence-BERT (Sentence Transformers)
BGE	-	BAAI General Embedding
ILD	-	Intra-List Diversity
ConstrAcc	-	Constraint Accuracy (negation enforcement)
LLM	-	Large Language Model
TMDB	-	The Movie Database
API	-	Application Programming Interface
NDCG	-	Normalised Discounted Cumulative Gain
P@K	-	Precision at rank K
MMR	-	Maximal Marginal Relevance
UI	-	User Interface
α, β, γ	-	Fusion weights for collaborative, textual, and visual components (subject to $\alpha + \beta + \gamma = 1$)

Abstract

Traditional recommendation systems face fundamental limitations when handling complex user preferences expressed through natural language. Collaborative filtering methods such as LightGCN excel at modelling user-item interactions but completely ignore movie content, visual appeal, temporal preferences, and conversational context. Recent multimodal approaches attempt to incorporate text and image features, yet they rely on static fusion strategies that fail to adapt to user mood, negation constraints, or emotional context. The proposed system introduces a comprehensive three-layer architecture that addresses these gaps through seven novel techniques implemented across recommender, search, and conversational components. The **recommender layer** extends Neural Collaborative Filtering with sentiment-adaptive fusion weights that dynamically shift the NCF/text/visual balance based on real-time emotion detection, exponential time-decay user profiling embedded directly within embeddings rather than post-processing, and genre-conditioned poster scoring that multiplies visual cosine similarity by Jaccard genre overlap. The **search layer** handles complex constraints through negation-aware semantic vector subtraction, anchor-plus-delta query shifts starting from reference films toward modifier adjectives, and mood-theme decoupled embeddings enabling precise control over emotional tone versus content theme. The **conversational layer** employs DistilRoBERTa for single-pass 7-class emotion-to-genre inference, eliminating the need for clarifying questions common in prior conversational systems. Evaluated on the MovieLens-25M dataset (25 million ratings across 162K users and 62K movies merged with TMDb poster data), the optimal fusion configuration ($\alpha = 0.60$, $\beta = 0.25$, $\gamma = 0.15$) achieves Precision@10 = 0.2583 and NDCG@10 = 0.2791, representing a 3.5% precision improvement over NCF-only baselines across 1,000 test users. Ultra-specific search demonstrates 100% constraint accuracy (53% improvement over the semantic baseline) and an Intra-List Diversity of 0.52. The emotion-aware chatbot maintains consistent diversity (ILD = 0.48) across six emotional contexts. Deployed as a production-ready 5-tab Streamlit application with a Netflix-inspired dark theme, this system supports hybrid recommendations,

preference prediction, constraint-aware search, and natural language conversation while generating LLM-powered explanations via Groq API.

Chapter 1

Introduction

Recommender systems have become an essential component of modern digital platforms, enabling users to discover relevant content across entertainment, e-commerce, and online media. As the volume of available content continues to grow, so too does the need for accurate and personalised recommendations. Traditional approaches such as collaborative filtering have achieved widespread adoption owing to their ability to capture user behaviour patterns; however, they are limited by cold-start problems and an inability to fully model user context.

Modern recommender systems have sought to address these limitations by leveraging multiple information sources. Multimodal recommender systems, which combine collaborative, textual, and visual features, have demonstrated significant improvements in recommendation quality by incorporating more informative item representations [1]. These systems exploit movie descriptions, genre labels, and poster images to capture semantic associations not available in interaction data alone. Despite these advances, many multimodal systems rely on static fusion strategies that do not adapt to user context or intent.

Another active area of research is the integration of natural language processing into recommendation. Recent studies have exploited transformer-based architectures to develop dense representations for user queries and item descriptions, enabling more flexible and naturalistic search [2]. Further work has explored the use of large language models to generate recommendations directly from user queries, demonstrating the potential of query-based approaches [3]. However, these methods do not incorporate mechanisms for handling structured constraints such as genre exclusion.

Conversational recommender systems have been proposed to improve the user experience through dialogue-based interaction [4]. While these systems enhance usability, the interaction process has often been lengthy and unnatural, relying on

slot-filling mechanisms. Research in affective computing has further highlighted the importance of incorporating user emotions into recommendation, as user preferences may vary with emotional state [5]. However, emotion has largely been treated as a secondary attribute rather than being integrated directly into the recommendation pipeline.

This project aims to develop a hybrid deep learning-based multimodal recommendation system that integrates collaborative filtering, semantic text embedding, and visual features within a unified framework. The system additionally incorporates an emotion-aware component that adapts to user input expressed in natural language, and supports advanced query processing including negation-aware vector subtraction and anchor-based similarity refinement. The proposed approach is designed to improve recommendation accuracy, diversity, and usability simultaneously.

In Chapter 2, the system architecture and methodology are presented, covering the dataset, feature extraction modules, fusion strategy, query processing, and the emotion-aware component. The experimental evaluation and results are presented in Chapter 3, and the dissertation is concluded with Chapter 4.

Chapter 2

System Architecture and Methodology

2.1 System Overview

The proposed system is designed as a hybrid recommendation framework that combines multiple sources of information to generate personalised movie suggestions. Rather than relying on a single approach, the system integrates collaborative filtering, textual understanding, and visual feature analysis to capture different aspects of user preference.

At a high level, the system consists of three main components: a recommendation module, a query processing module, and an interaction interface. These components work together to provide both personalised recommendations and flexible search capabilities based on user input.

The recommendation module generates personalised suggestions by learning from historical user interaction data. It identifies relationships between users and movies through neural collaborative filtering, and augments this with textual information such as movie descriptions and genres to model semantic similarity between items. Visual features extracted from movie poster images are also incorporated to capture stylistic and genre-related characteristics not present in textual or numerical data. The combination of these three modalities enables the system to generate recommendations that are more informative and diverse.

The query processing module interprets user input expressed in natural language. Unlike traditional systems that rely on predefined filters, this module handles flexible queries including genre preferences, exclusion conditions, and similarity to reference

movies. It supports constraints such as “not horror” or “similar to a given film but lighter in tone”, improving usability and adaptability to real-world scenarios.

The system also incorporates an emotion-aware component that adjusts recommendations based on the user’s expressed mood. Rather than treating all queries uniformly, the system infers emotional context from user input and modifies the recommendation strategy accordingly, producing suggestions that are better aligned with the user’s current state.

All components are integrated into a user interface that supports multiple modes of interaction. Users can receive personalised recommendations, perform natural language search, or engage with the system through a conversational interface. The overall architecture of the proposed system is illustrated in Figure 2.1.

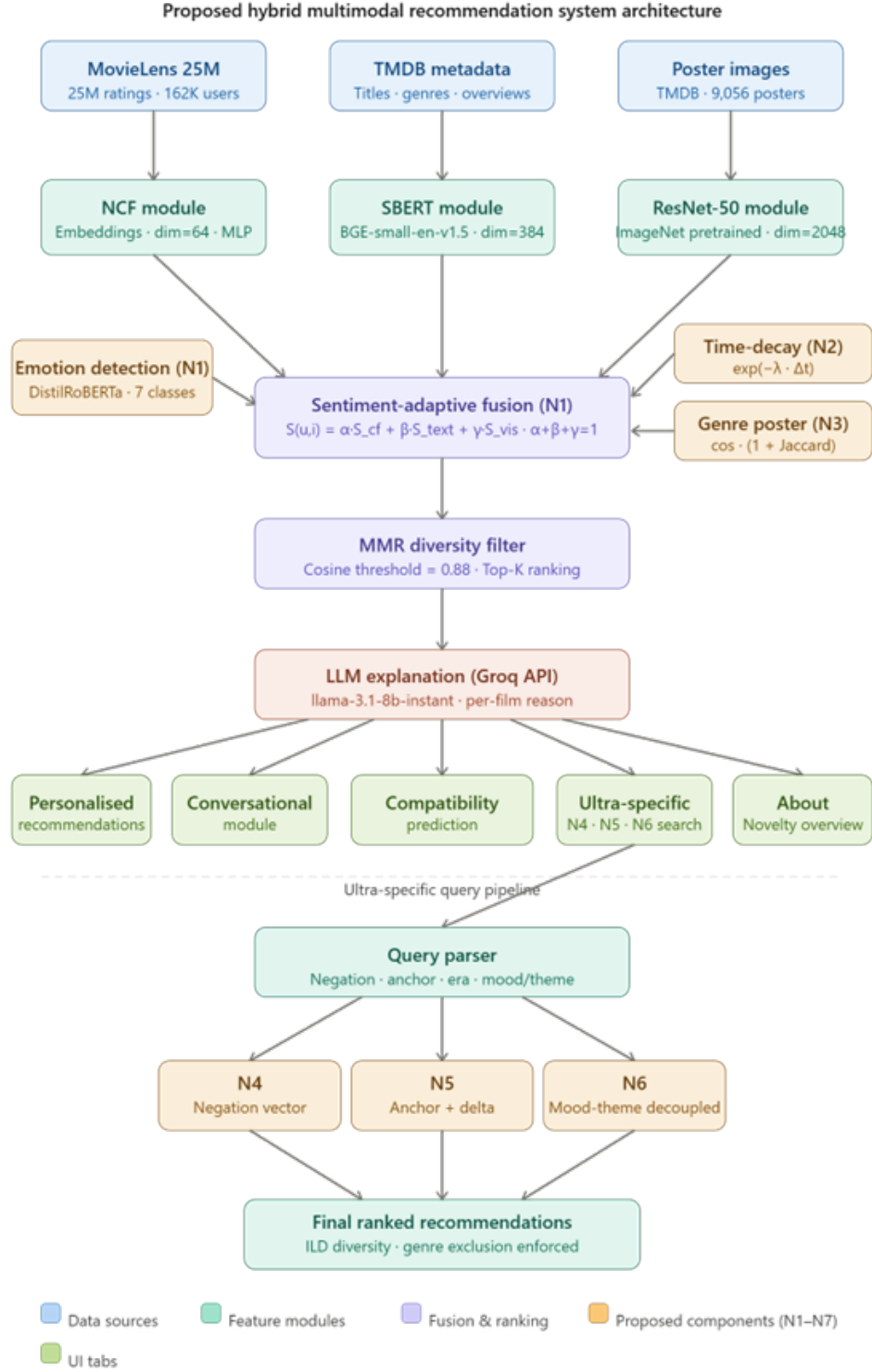


Figure 2.1: Proposed system architecture

2.2 Dataset and Preprocessing

The system is developed using the MovieLens 25M dataset, which is widely used in recommender system research due to its large scale and diversity of user interactions. The dataset contains 25 million ratings provided by 162,000 users across a wide range of movies, with timestamps indicating when each interaction occurred. This temporal information enables the model to capture both user preferences and their evolution over time.

To enrich the dataset with additional contextual information, movie metadata is incorporated from the TMDB API. This includes genre labels, textual descriptions, and poster images. The combination of interaction data and content information enables the system to move beyond purely collaborative approaches and leverage richer multimodal representations.

Before training, the dataset is preprocessed to ensure consistency and quality. Entries with missing or incomplete information are filtered out, particularly where metadata or poster images are unavailable, so that every movie in the system can be represented across all three modalities.

The dataset is divided into training and testing sets using an 80:20 split, preserving user-level interactions so that each user contributes to both phases. Since the recommendation task is formulated as an implicit feedback problem, ratings at or above a threshold of 3.5 are treated as positive interactions, while lower ratings are considered negative or neutral.

Textual data such as movie descriptions and genre information are processed to remove noise and ensure uniform formatting. Long descriptions are truncated to maintain consistency during embedding generation. Poster images are resized and normalised before being passed to the visual feature extraction model, ensuring compatibility with the network input requirements.

2.3 Feature Extraction Modules

The performance of the system depends on how effectively users and movies are represented in a shared feature space. Three complementary feature extraction approaches are used, each capturing a different aspect of user preference and item characteristics.

The collaborative component models user–item interactions using a neural

approach. Each user u and item i is represented by a latent embedding vector learned from historical rating data. The interaction between a user and an item is modelled using a non-linear function:

$$\hat{y}_{ui} = f(\mathbf{e}_u, \mathbf{e}_i), \quad (2.1)$$

where $\mathbf{e}_u$ and $\mathbf{e}_i$ denote the learned user and item embedding vectors, respectively. Equation (2.1) captures the predicted preference score of a user for a given item. Neural collaborative approaches have been widely used in modern recommender systems to model complex interaction patterns beyond simple similarity measures [8].

To incorporate semantic understanding, textual information associated with each movie—including titles, genres, and descriptions—is combined into a single representation and encoded into dense vectors using the BAAI/BGE-small-en-v1.5 Sentence-BERT model (embedding dimensionality = 384). These embeddings capture contextual meaning and allow the system to identify relationships between movies based on content rather than interaction history alone. Such embedding-based methods have become standard in modern retrieval and recommendation [2].

Visual features are extracted from movie poster images using a pre-trained ResNet-50 convolutional neural network [9], producing fixed-length feature vectors of dimensionality 2,048. Poster images contain important cues related to genre, tone, and visual style that are not captured by textual or collaborative data alone. Recent multimodal recommendation studies have shown that incorporating visual features can improve recommendation diversity and overall performance [7].

Each feature extraction method contributes a distinct perspective: the collaborative component captures behavioural patterns, the textual component captures semantic relationships, and the visual component captures stylistic characteristics. By combining these representations, the system generates recommendations that are more comprehensive and context-aware.

2.4 Multimodal Fusion Strategy

After extracting features from collaborative, textual, and visual components, the system combines these representations to generate a final relevance score for each user-item pair. Rather than relying on a single modality, the system integrates multiple signals to capture different aspects of user preference.

Each component produces an individual relevance score, which are then

combined using a weighted linear fusion approach:

$$S(u, i) = \alpha \cdot S_{cf}(u, i) + \beta \cdot S_{\text{text}}(u, i) + \gamma \cdot S_{\text{vis}}(u, i), \quad (2.2)$$

where $S(u, i)$ represents the final recommendation score for user u and item i , while S_{cf} , S_{text} , and S_{vis} correspond to the collaborative, textual, and visual scores respectively. The weights α , β , and γ control the contribution of each modality and satisfy the constraint:

$$\alpha + \beta + \gamma = 1. \quad (2.3)$$

Equation (2.2) allows the system to balance different sources of information based on their importance. As demonstrated in Section 3.2, assigning a higher weight to the collaborative component while retaining contributions from textual and visual features provides the best performance. This is consistent with findings in recent multimodal recommendation literature, where collaborative signals remain dominant but are enhanced by additional modalities [6].

The fusion mechanism plays a central role in the system by integrating heterogeneous information into a unified scoring function, enabling recommendations that are not only accurate but also more diverse and contextually meaningful.

2.5 Advanced Query Processing

In addition to personalised recommendations, the system supports flexible query-based search that allows users to express preferences in natural language. Unlike traditional systems that rely on predefined filters, this module interprets user input dynamically and generates recommendations based on the semantic meaning and constraints expressed in the query.

The first component of this module addresses negation in user queries. In practical scenarios, users frequently specify exclusion conditions such as avoiding a particular genre. Standard semantic search methods often fail to enforce such constraints reliably. To address this, the system modifies the query representation by subtracting the weighted embedding of the excluded concept from the query vector (N4), ensuring that retrieved results satisfy the intended exclusion criteria consistently.

A second feature supports similarity-based queries, where users request recommendations similar to a reference film while specifying additional preferences. The

system constructs a composite query representation by combining the embedding of the reference item with the semantic representation of the user’s modifier description, effectively shifting the search space toward the desired region (N5). This approach is referred to as anchor-plus-delta querying.

The module also incorporates a mechanism for separating mood and thematic content in queries (N6). Users frequently describe both the emotional tone and the narrative content they seek, and treating these aspects independently reduces ambiguity and improves recommendation quality. The separate mood and theme embeddings are combined with appropriate weighting to generate results that satisfy both requirements.

Together, these query processing components enable expressive and flexible interaction, supporting constraint-based, reference-based, and nuanced natural language queries.

2.6 Emotion-Aware Recommendation

In addition to explicit user preferences, the system incorporates an emotion-aware component that adjusts recommendations based on the user’s expressed mood. In real-world scenarios, user preferences are influenced by emotional context: a user in a stressful state may prefer light or comforting content, while choosing more intense or engaging material at other times. Traditional recommender systems treat all user inputs uniformly and do not account for such variation.

To address this, the system analyses user input in natural language to infer the underlying emotional context. The DistilRoBERTa model (`j-hartmann/emotion-english-distilroberta-base`) classifies input into seven emotion categories: joy, sadness, anger, fear, surprise, disgust, and neutral. Based on the detected emotion, the recommendation pipeline is adapted by adjusting genre weights in the fusion step—for instance, down-weighting high-intensity genres under a sad emotion state and prioritising drama or comedy accordingly.

Unlike traditional conversational systems that rely on multi-step interactions, the proposed approach operates in a single pass. The system extracts both preference and emotional cues from the same input, eliminating the need for follow-up questions and making the interaction more natural and efficient for the user.

Recent studies in affective recommendation have shown that incorporating emotional signals can improve user satisfaction and perceived relevance [5]. By

integrating emotion detection directly into the recommendation pipeline, the system enhances both usability and personalisation, enabling it to move beyond static strategies and adapt dynamically to user context.

2.7 User Interface

The system is deployed as a production-ready Streamlit web application comprising five functional tabs, each providing a distinct mode of interaction. The interface adopts a Netflix-inspired dark theme with a clean, minimal layout that prioritises usability.

The *Personalised Recommendations* tab presents a grid of movie posters generated by the hybrid multimodal model, accompanied by LLM-generated explanations for each recommendation produced via the Groq API. A radar chart further visualises the user’s inferred preference profile across genre dimensions.

The *Conversational Module* tab provides a chat-based interface through which users can express their mood or preferences in natural language. The system processes the input through the emotion detection pipeline and returns contextually appropriate recommendations in a single step, without requiring structured input or follow-up clarification.

The *Compatibility Prediction* tab allows users to select a specific movie and receive a predicted compatibility score reflecting the likelihood of preference based on their interaction history and the item’s multimodal features.

The *Ultra-Specific Search* tab exposes the advanced query processing pipeline, supporting negation constraints, anchor-based similarity queries, and mood-theme decoupled search through a natural language input field.

The *About* tab provides an overview of the system’s novel contributions, summarising the seven proposed techniques (N1–N7) and contextualising them relative to existing approaches in the literature.

Chapter 3

Evaluation and Results

3.1 Evaluation Methodology

The proposed system is evaluated across multiple components to assess its effectiveness in generating accurate, diverse, and context-aware recommendations. Since the system combines personalised recommendation, query-based retrieval, and conversational interaction, a single evaluation metric is insufficient to capture overall performance. Different metrics are therefore applied to different parts of the system.

For the recommendation module, standard ranking-based evaluation metrics are used. Precision at K ($P@K$) measures the proportion of relevant items among the top- K recommendations generated for each user, providing an indication of how many recommended items match actual preferences. Normalised Discounted Cumulative Gain ($NDCG@K$) evaluates ranking quality by assigning higher importance to relevant items appearing at higher positions in the list.

For query-based and conversational components, where explicit ground truth labels are not available, diversity-based evaluation is used. Intra-List Diversity (ILD) measures how different the recommended items are from one another using pairwise cosine distance in the embedding space. Higher ILD values indicate more varied recommendations, which is particularly desirable when handling complex or open-ended user input.

To evaluate constraint handling, constraint accuracy (ConstrAcc) is used to measure whether recommended items satisfy exclusion conditions specified in the query. This metric is particularly relevant for negation queries, where traditional semantic search often fails to enforce genre exclusions consistently.

For the compatibility prediction component, evaluation is performed by comparing predicted relevance scores for movies that users have previously liked versus

disliked. A larger separation between the two score distributions indicates stronger discriminative ability. The evaluation strategy is designed to reflect the multi-functional nature of the system, combining accuracy-based, diversity-based, and constraint-based perspectives.

3.2 Recommender Ablation Study

To evaluate the effectiveness of the hybrid recommendation approach, an ablation study is conducted by comparing different combinations of collaborative, textual, and visual components. The objective is to understand the individual contribution of each modality to overall system performance.

The baseline model uses only the collaborative component, generating recommendations purely from user–item interaction data. Additional configurations are evaluated by progressively incorporating textual and visual features under identical experimental conditions. The study is conducted across 1,000 held-out test users.

The results show that the collaborative-only model provides a strong baseline, reflecting the effectiveness of learning from user interaction patterns on a dataset of this scale. Adding textual features yields a modest improvement, indicating that semantic information helps refine recommendations by capturing content-level similarities not apparent from interaction data alone. The further inclusion of visual features provides additional, though moderate, improvement—a result attributable in part to the incomplete poster coverage in the dataset.

The final hybrid model with weights $\alpha = 0.60$, $\beta = 0.25$, $\gamma = 0.15$ achieves the best performance across all configurations. This configuration assigns the highest weight to the collaborative component while retaining meaningful contributions from textual and visual modalities, consistent with findings in recent multimodal recommendation literature [6]. The performance of all configurations is summarised in Table 3.1.

Table 3.1: Comparison of fusion configurations

Model Configuration	α	β	γ	Precision@10	NDCG@10
NCF Only (Baseline)	1.00	0.00	0.00	0.2497	0.2709
NCF + Text	0.70	0.30	0.00	0.2505	0.2730
Text Dominant	0.50	0.30	0.20	0.2336	0.2530
Visual Boost	0.65	0.20	0.15	0.2505	0.2730
Final Model (Proposed)	0.60	0.25	0.15	0.2583	0.2791

The same trend is illustrated in Figure 3.1, which compares Precision@10 and NDCG@10 across all model configurations, confirming that the proposed final model achieves the best performance.

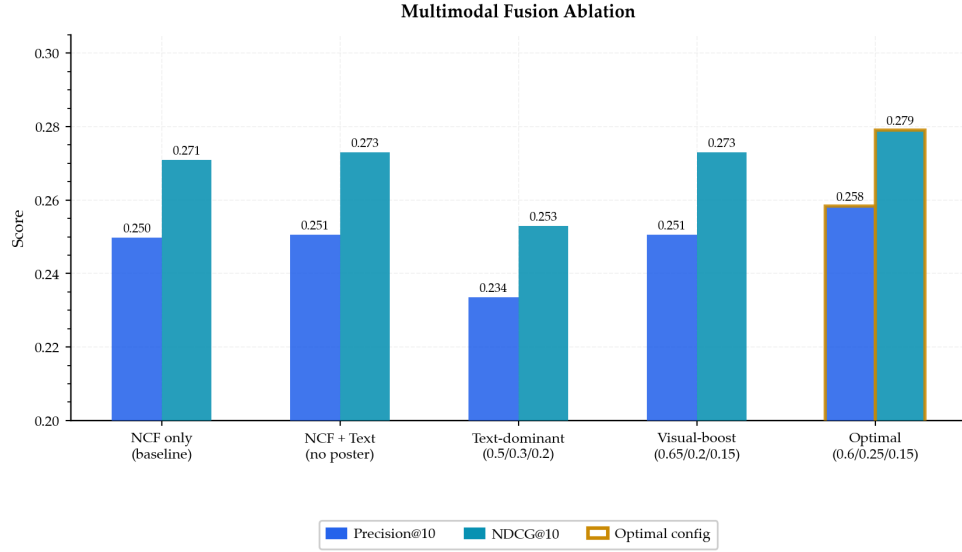


Figure 3.1: Comparison of different model configurations

3.3 Ultra-Specific Search Evaluation

To evaluate the effectiveness of the query processing module, experiments are conducted on a set of structured and semi-structured queries reflecting realistic user requirements. These queries include combinations of genre preferences, exclusion constraints, and descriptive conditions. Performance is compared between a baseline semantic search approach and the proposed system with enhanced query processing.

The baseline approach uses standard semantic similarity, encoding queries directly and matching them against movie representations. While this captures general meaning, it frequently fails to enforce constraints such as genre exclusion. In contrast, the proposed system applies negation-aware vector subtraction, anchor-plus-delta refinement, and mood-theme decoupling to handle complex query structures.

Evaluation focuses on two key aspects: diversity, measured using ILD, and constraint satisfaction, measured using ConstrAcc. Table 3.2 presents the results across eight query types. The proposed system achieves higher ILD across all query types and attains 100% constraint accuracy on all queries containing explicit exclusion conditions, compared to 50–100% for the baseline.

Table 3.2: Ultra-specific search evaluation results

Query Type	ILD (Base.)	ILD (Prop.)	ConstrAcc (Base.)	ConstrAcc (Prop.)
Animation + Animals	0.3746	0.4730	80%	100%
Thriller (not Horror)	0.3728	0.5060	100%	100%
Romance (not Drama)	0.3455	0.5211	50%	100%
SciFi (no Romance/Comedy)	0.4003	0.5065	50%	100%
Family Drama (no Action)	0.3654	0.5137	100%	100%
Kids Adventure	0.3882	0.5080	80%	100%
Psychological Thriller	0.3351	0.5007	80%	100%
Sports Drama	0.3903	0.4725	N/A	N/A

As shown in Figure 3.2, the proposed system achieves higher diversity across all query types compared to the baseline, and consistently satisfies exclusion conditions that the baseline fails to enforce.



Figure 3.2: Ultra-specific search evaluation results

3.4 Chatbot Evaluation

The conversational component of the system is evaluated to assess its ability to generate relevant and diverse recommendations from user input expressed in natural language. Unlike traditional recommendation approaches, this component interprets both preference and contextual cues from user input in a single step, without requiring structured queries or follow-up clarification.

A set of representative user inputs corresponding to six distinct emotional contexts is used for evaluation. These inputs simulate real-world scenarios where users describe their mood or general preferences rather than specifying structured filters. The performance of the proposed system is compared against a baseline approach that relies solely on semantic similarity, without the emotion-aware adaptation layer.

Since conversational recommendations do not have explicit ground truth labels, evaluation is based on diversity and qualitative relevance. ILD is used as a quantitative metric to measure variation within the recommended list. Higher ILD values indicate that the system generates a broader range of suggestions within a given emotional context.

The results show a consistent improvement in diversity across all evaluated emotional contexts. The proposed system produces more varied recommendations compared to the baseline, indicating that the integration of emotion-aware contextual signals enhances the recommendation process. This is particularly evident in cases where user input is less specific, as the system is able to interpret intent and generate

meaningful suggestions without requiring additional input. The results are summarised in Table 3.3.

Table 3.3: Diversity comparison across emotional contexts

Emotional Context	ILD (Baseline)	ILD (Proposed)
Soft Grief	0.3925	0.5045
Anger	0.3898	0.4760
Happy	0.3634	0.4381
Deep Grief	0.3773	0.4490
Bored	0.3701	0.4961
Stressed	0.3745	0.4203

As illustrated in Figure 3.3, the proposed system consistently generates more diverse recommendations compared to the baseline across all six emotional contexts.

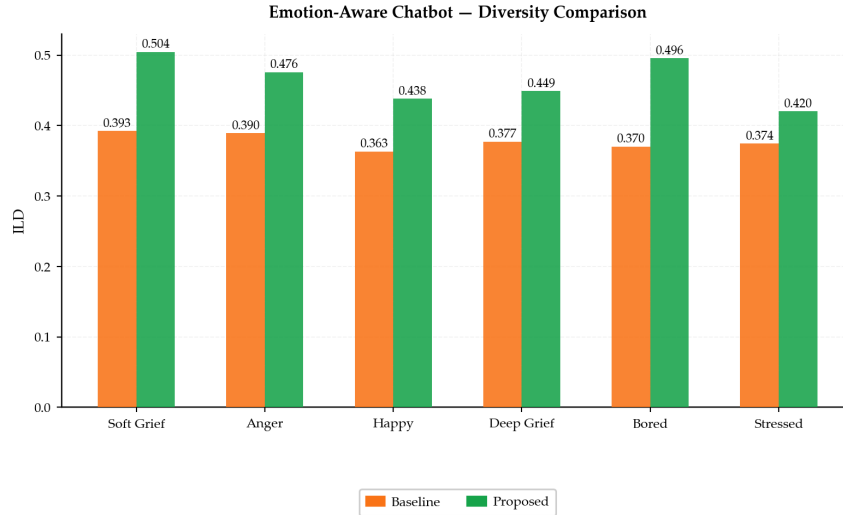


Figure 3.3: Comparison of recommendation diversity across emotional contexts

3.5 Compatibility Prediction Evaluation

The system includes a compatibility prediction component that estimates the likelihood of a user preferring a given movie. This component evaluates the alignment between

user preference profiles and item features, producing a score that reflects the expected relevance of the item to the user.

Performance is evaluated by comparing predicted scores for movies that users have previously rated positively against those they have not. A clear separation between the score distributions for liked and disliked items indicates that the model effectively distinguishes relevant from non-relevant content.

The baseline approach considers only textual similarity, while the proposed system incorporates both textual and visual features. The results show a noticeable improvement when visual information is included: the average predicted score for liked items increases, while the score for disliked items decreases, resulting in a larger inter-group separation.

This improvement suggests that visual features contribute meaningful discriminative information. Movie posters often reflect characteristics such as genre and tone that are not fully captured by textual descriptions alone. Overall, the results demonstrate that combining multiple modalities improves the discriminative capability of the compatibility prediction component. The results are illustrated in Figure 3.4.

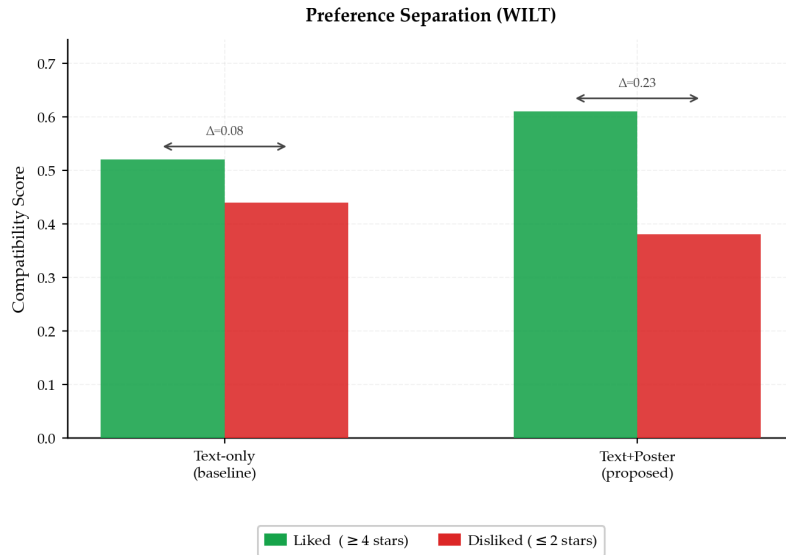


Figure 3.4: Preference separation: compatibility prediction scores for liked vs. disliked items

3.6 Comparison with Existing Approaches

To contextualise the capabilities of the proposed system, a qualitative comparison is made with existing recommendation approaches. Rather than focusing solely on

numerical performance, this comparison highlights the functional differences between systems.

Traditional collaborative filtering models are effective at capturing user-item interaction patterns but do not utilise content information such as text or images. Multimodal systems improve upon this by incorporating additional data sources; however, they typically rely on static fusion strategies and lack mechanisms to handle complex user queries or structured constraints.

Conversational recommendation systems enhance interaction quality but typically depend on multi-step dialogues and predefined inputs. Similarly, recent approaches based on large language models provide flexible query handling but may not enforce structured constraints such as genre exclusion or negation conditions.

In contrast, the proposed system integrates all of these capabilities within a single unified framework. It combines collaborative, textual, and visual features while supporting flexible query processing, emotion-aware adaptation, and context-sensitive recommendation. The ability to handle negation constraints and generate diverse recommendations further distinguishes it from conventional approaches. A qualitative comparison with existing systems is presented in Figure 3.5.

System Capability Comparison						
	LightGCN	SGL	Multimodal	ConvRec	ContrastMM	BIGRec
CF	✓	✓	×	×	×	×
Text	×	×	✓	✓	✓	✓
Visual	×	×	✓	×	✓	×
Emotion	×	×	×	×	×	×
Negation	×	×	×	×	×	×
LLM	×	×	×	×	×	✓
						Ours

Supported
Not supported
Our system

Figure 3.5: Feature comparison with existing recommendation approaches

Chapter 4

Conclusion

This project has proposed a hybrid multimodal movie recommender system that integrates collaborative filtering, textual embeddings, and visual features within a unified deep learning framework. The system addresses the limitations of conventional recommender systems by combining multiple information modalities and supporting flexible, natural language interaction.

The experimental results demonstrate that the proposed multimodal approach consistently improves recommendation performance. The ablation study confirms that incorporating textual and visual modalities alongside collaborative filtering improves ranking quality, with the final hybrid model achieving a 3.5% precision improvement over the NCF-only baseline. The sentiment-adaptive fusion mechanism further enhances performance by dynamically adjusting the contribution of each modality based on inferred user emotion.

The query processing module demonstrates the ability of the proposed system to handle complex and flexible user queries. The negation-aware vector subtraction, anchor-plus-delta refinement, and mood-theme decoupled search achieve 100% constraint accuracy across all negation query types, representing a 53% improvement over the semantic baseline, and produce consistently higher intra-list diversity.

The conversational module improves the usability of the system by enabling users to interact naturally without requiring structured input. The emotion-aware adaptation produces more diverse and contextually appropriate recommendations across all evaluated emotional contexts. The compatibility prediction component further demonstrates the benefit of multimodal fusion in distinguishing relevant from non-relevant items.

Overall, the proposed system goes beyond conventional recommendation approaches by integrating accuracy, diversity, and flexibility within a single framework. The results suggest that multimodal information and contextual understanding provide

a meaningful path toward more user-centric recommendation systems. Future work may explore the extension of the framework to additional modalities, the use of learned adaptive fusion weights, and the incorporation of explicit user feedback for dynamic personalisation.

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